Mohammed Mahdi Jahangiri

Forward-Deployed AI Engineer

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AI engineer with 2+ years of production delivery and an MSc in AI from Cardiff University. For my dissertation, fine-tuned and open-sourced a 2B model that beat a 26B baseline on task accuracy (53.4% vs 29.0%) while cutting unsafe outputs from 25 to 5 per 500 trials; Now shipped on GitHub, Hugging Face and Nexus Mods. Took sprint success from 40% to 90% leading a 3–6 person engineering team at Landell Games.

Technical Skills

Core Languages: Python, C++, C#
ML & AI: PyTorch, Hugging Face Transformers, LLM fine-tuning, QLoRA, evaluation pipelines, prompt engineering, NLP, RAG
Engineering: FastAPI, REST APIs, SQL, JSON, Git, CI/CD, Agile/Scrum, code review, cross-functional coordination

Experience, Projects & Publications

MSc Dissertation — Local LLM NPC, Skyrim

Python, QLoRA, Gemma 4 (E2B), Papyrus

- Engineered a **Python** orchestration layer for a Skyrim mod, bridging deterministic **Papyrus** scripts with a local LLM via a **JSON-schema** validator and fallback state machine that **blocks hallucinated game actions** before they reach the engine
- Fine-tuned a **Gemma 4 (E2B)** model via **QLoRA** 4-bit quantisation on consumer hardware, lifting task-routing accuracy from **19.0% to 53.4%**, beating a 26B model's 29.0% zero-shot at **sub-second latency**
- Built an **automated evaluation harness** (100 frozen cases × 5 seeds = **1,500 trials**) measuring accuracy, latency and JSON compliance across the base 2B, 26B and fine-tuned models
- Ran a dedicated 500-trial safety benchmark, cutting unsafe/hallucinated outputs from **25 to 5 cases**; open-sourced the model, mod and code (MIT) on **GitHub**, **Hugging Face** and **Nexus Mods**

R&D Technical Artist (Contract) — Awkward Studios / Media Cymru

Apr 2026 – Jun 2026

Cardiff — R&D for CULTVR Lab: 360° dome-projection game-show installation.

- Delivered **forward-deployed R&D across four organisations**: Awkward Studios, Small Important Things Studios, venue partner **CULTVR** and sponsor **Media Cymru**, ending in a **live public QA session** on the installation's test day
- Bridged gaps in open-source dome tooling where **no off-the-shelf UI spec existed**, writing my own **Unity (C#)** fix for UI text distortion on the 360° dome
- Structured the game-show's mini-game flow as modular state transitions, keeping sequence logic separate from the UI
- Collaborated through a GitHub pull-request workflow in a small cross-disciplinary R&D team

Engineering Team Coordinator — Landell Games

Feb 2025 – Sep 2026

Remote / Sweden — Promoted from Engineering Intern (Dec 2024 – Jan 2025). Team lead on a live-service software project.

- Owned **weekly build pipelines** and mediated cross-functional concerns between **5 department leads**
- Coordinated a **3–6 person team**, turning priorities into sprint tasks and raising sprint success from **40% to 90%**
- Profiled and optimised system performance bottlenecks, improving frame time by **10–30 FPS** on target hardware
- Conducted **code reviews** and enforced clean **Git/Plastic** branch hygiene across a remote team
- Implemented data-driven configuration and gameplay state-machine logic in **C#**, supporting designer-led balance tweaks

Publication — ICHORA 2024

LLM and Pursuit Learning Automata for NPC Dialogues

- Published and presented a resource-efficient local LLM pipeline for dynamic NPC dialogue, applying **pursuit learning automata** to balance satisfaction against compute cost

Solo-Shipped Spatial App — Meta Quest Store

Unity, C#, Meta Quest SDK

- Owned the **full pipeline solo**: build, compliance review and **store approval**, then iterated post-launch on real user telemetry

AI-Driven NPC Interaction Prototype

Unreal Engine 5, C++, Python, FastAPI

- Built a local **Python FastAPI** server bridging real-time microphone input to a **C++** application via a custom REST API
- Integrated **Whisper**, **Wit.ai** and local **Hugging Face** models for real-time voice and emotion detection, streaming predicted signals into **NPC** behaviour-tree logic
- Trained the emotion model and configured Wit.ai intents using custom voice recordings

Education

MSc Artificial Intelligence — Cardiff University

2025–2026

Dissertation completed: locally hosted LLM-driven NPC deployed alongside Skyrim.

- Computer Vision & NLP: bounding-box localisation via **RF-DETR** on **VOC2012**; NLP pipelines with **pandas/scikit-learn** for data gathering, labelling, and intent/emotion analysis
- **AI Safety Founding Fellow (Spring 2026):** Completed fundamentals fellowship on catastrophic AI risks, model evaluation and AI governance

BEng Computer Engineering — Azad University, Tehran

2020–2025

- Graduated with **First-Class Honours**; core study: **Data Structures & Algorithms**, **Advanced Programming (C++)**, **SQL** and database design, Computational Architecture and Networks

Awards & Interests

DiffTrailblazers — Cardiff University: Won 1st place, receiving a **£1,500** prize for a practical innovation project.

Interests: Judo (green belt) | Mandarin Chinese | Playing with AI agents and automation